

23) Suppose a 3×5 coefficient matrix for a system has three pivot columns. Is the system consistent? Why or why not?

Soln: By definition, the pivot position of a matrix is that position which corresponds to the leading 1 of one of the rows of the matrix in reduced row echelon form. The pivot columns are those containing at least one pivot position.

From condition 2 of the defn. of reduced row echelon form of a matrix, we know that each leading entry of a row is in a column to the right of the column of all such leading entries of the rows above it. $\therefore$ Each pivot column has exactly one pivot position.

Since there are 3 pivot columns, hence there are 3 pivots. There are 3 rows of the matrix. $\therefore$ Each row has a pivot. By defn. of a ~~pivot~~ leading entry, there are no ~~non~~-zero rows of the matrix since each row has at least one non-zero element.

By Theorem 2, a system is inconsistent if and only if at least one row of the augmented matrix in echelon form is of the form $[0 \ 0 \ \dots \ 0 \ b]$, where $b \neq 0$. $\therefore$ At least one row of the coefficient matrix must be zero.

$\therefore$ Our given matrix corresponds a consistent system

24) Suppose a system of linear equations has a 3×5 augmented matrix where fifth column is a pivot column. Is the system consistent? Why (or why not)?

Soln: By defn, a pivot position of a matrix represents that position which contains a leading 1 in the reduced row echelon form. A pivot column contains at least one pivot. By defn. of leading entry, in the reduced row echelon form, the row corresponding to that pivot is $[0 \ 0 \ 0 \ 0 \ 1]$. By Theorem 2, the given system is inconsistent since the corresponding augmented matrix contains at least one row of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$, $b \neq 0$.